

24.5.2022 (E)

**K. J. Somaiya College of Engineering, Mumbai-77**  
(Autonomous College Affiliated to University of Mumbai)

**End Semester Examinations**  
May-June 2022

Max. Marks: 50

Class: TY

Name of the Course: Artificial Intelligence

Course Code: 2UCC603

Duration: 2 hours

Semester: VI

Branch: Computer

**Instructions:**

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

PwD

| Question No. | SECTION A                                                                                                                                                                                                                                               | Marks               |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Q1(a)        | State and explain the "AI Problems". Brief on Mundane, Formal and Expert tasks.<br><br>OR<br>(i) List 5 things AI cannot do yet.<br>(ii) List 5 applications of AI in daily life.                                                                       | 10M<br><br>5M<br>5M |
| Q1(b)        | Describe each of the components of the Utility based agent. Support with a neat diagram.<br>Comment on the example of a Mars lander on the surface of mars with obstacles in its way and how would it be different in a Goal based agent with learning. | 3M<br>3M<br>4M      |
| Q2(a)        | What are Production rules? <u>OR Q.2 A Below</u><br>List the production rules for Water Jug Problem                                                                                                                                                     | 2M<br>8M            |
| Q2(b)        | Define straight line distance as a heuristic.<br>Distinguish between AO* Search Algorithm and A* search algorithm.<br>Differentiate between BFS and DFS.                                                                                                | 02M<br>04M<br>04M   |
| Q3.1         | Artificial Intelligence is about _____.<br><br>1. Playing a game on Computer<br><br>2. Making a machine Intelligent<br><br>3. Programming on Machine with your Own Intelligence<br><br>4. Putting your intelligence in Machine                          | 1M                  |
| Q3.2         | Who is known as the "Father of AI"?<br><br>1. Fisher Ada<br>2. Alan Turing<br>3. John McCarthy<br>4. Allen Newell                                                                                                                                       | 1M                  |

Q. 2 A. Explain steps in genetic algorithms with a suitable example. 10M.  
(2M for each step.)

|      |                                                                                                                                                                                                                                                                                            |    |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Q3.3 | <p>Among the given options, which search algorithm requires less memory?</p> <ol style="list-style-type: none"> <li>1. Optimal Search</li> <li>2. Depth First Search</li> <li>3. Breadth-First Search</li> <li>4. Linear Search</li> </ol>                                                 | 1M |
| Q3.4 | <p>A technique that was developed to determine whether a machine could or could not demonstrate the artificial intelligence known as the ____</p> <ol style="list-style-type: none"> <li>1. Boolean Algebra</li> <li>2. Turing Test</li> <li>3. Logarithm</li> <li>4. Algorithm</li> </ol> | 1M |
| Q3.5 | <p>Which algorithm is used in the Game tree to make decisions of Win/Lose?</p> <ol style="list-style-type: none"> <li>1. Heuristic Search Algorithm</li> <li>2. DFS/BFS algorithm</li> <li>3. Greedy Search Algorithm</li> <li>4. Min/Max algorithm</li> </ol>                             | 1M |
| Q3.6 | <p>An AI agent perceives and acts upon the environment using ____.</p> <ol style="list-style-type: none"> <li>1. Sensors</li> <li>2. Perceiver</li> <li>3. Actuators</li> <li>4. Sensors and Perceivers</li> </ol>                                                                         | 1M |
| Q3.7 | <p>Which agent deals with the happy and unhappy state?</p> <ol style="list-style-type: none"> <li>1. Utility-based agent</li> <li>2. Model-based agent</li> <li>3. Goal-based Agent</li> <li>4. Learning Agent</li> </ol>                                                                  | 1M |
| Q3.8 | <p>Which of the following is not an application of artificial intelligence?</p> <ol style="list-style-type: none"> <li>1. Computer Vision</li> <li>2. Natural Language Processing</li> <li>3. Database Management System</li> <li>4. Digital Assistants</li> </ol>                         | 1M |
| Q3.9 | <p>What is the work of Task Environment and Rational Agents?</p> <ol style="list-style-type: none"> <li>1. Problem and Solution</li> <li>2. Solution and Problem</li> <li>3. Observation and Problem</li> <li>4. Observation and Solution</li> </ol>                                       | 1M |

|              |                                                                                                                                                                                                |           |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Q3.10</b> | <b>Machines that try to imitate human intuition while handling vague information lie in the field of AI called?</b><br>1.Functional Logic<br>2.Fuzzy Logic<br>3.Boolean Logic<br>4.Human Logic | <b>1M</b> |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|



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**Max. Marks: 50**

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Name of the Course: Artificial Intelligence

Course Code: 2UCC603

**Duration: 2 hours**

Semester: VI

Branch: Computer

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PWD

|        | <b>SECTION B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Q 4(a) | Solve using CSP<br>TWO+TWO=FOUR                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 10M        |
|        | OR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |
| Q 4(a) | Explain the steps of knowledge engineering process for a detective agent. Give sample output for every step.                                                                                                                                                                                                                                                                                                                                                                      | 10M        |
| Q 4(b) | Translate the following set of sentences into FOL. Use backward chaining on the goal- Joel is at museum.<br><br>a. Joel is a bulldog.<br>b. John is Joel's master.<br>c. The day is Saturday<br>d. It is cold on Saturday.<br>e. Joel is trained.<br>f. Trained bulldogs are good dogs.<br>g. If a dog is a good dog and has a master and at some place, then he will be with his master.<br>h. If the day is Saturday and the day is cold, John is at the museum.<br><i>Give</i> | 10M        |
| Q 5(a) | STRIPS solution for the problem of changing flat tire with a spare one in car trunk.<br>Clearly specify initial state, goal(s), actions and their pre-post conditions                                                                                                                                                                                                                                                                                                             | 10M        |
| Q 5(b) | Differentiate between Supervised learning, Unsupervised learning, Semi supervised learning and reinforcement learning.<br><br>OR<br>List and explain the steps to build an NLP                                                                                                                                                                                                                                                                                                    | 10M<br>10M |

|       |                                                                                                                                                                                                                                                                                                                                                                       |    |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Q 6.1 | <p>Which algorithm is used in the Game tree to make decisions of Win/Lose?</p> <ol style="list-style-type: none"> <li>1. Heuristic Search Algorithm</li> <li>2. DFS/BFS algorithm</li> <li>3. Greedy Search Algorithm</li> <li>4. Min/Max algorithm</li> </ol>                                                                                                        | 1M |
| Q6.2  | <p>In the Wumpus World Problem, the reason for the uncertainty is that the agent's sensor gives only__</p> <ol style="list-style-type: none"> <li>1. Full &amp; Global information</li> <li>2. Partial &amp; Global Information</li> <li>3. Full &amp; local information</li> <li>4. Partial &amp; local Information</li> </ol>                                       | 1M |
| Q 6.3 | <p>The maximum depth to which the alpha-beta pruning can be applied.</p> <ol style="list-style-type: none"> <li>1. Eight states</li> <li>2. Six states</li> <li>3. Ten states</li> <li>4. Any depth</li> </ol>                                                                                                                                                        | 1M |
| Q 6.4 | <p>Face Recognition system is based on which type of approach?</p> <ol style="list-style-type: none"> <li>a) Weak AI approach</li> <li>b) Applied AI approach</li> <li>c) Cognitive AI approach</li> <li>d) Strong AI approach</li> </ol>                                                                                                                             | 1M |
| Q6..5 | <p>What does the Bayesian network provide?</p> <ol style="list-style-type: none"> <li>a) Partial description of the domain</li> <li>b) Complete description of the problem</li> <li>c) Complete description of the domain</li> <li>d) None of the mentioned</li> </ol>                                                                                                | 1M |
| Q6..6 | <p>Which of the following is an application of Artificial Intelligence?</p> <ol style="list-style-type: none"> <li>a) It helps to exploit vulnerabilities to secure the firm</li> <li>b) Language understanding and problem-solving (Text analytics and NLP)</li> <li>c) Easy to create a website</li> <li>d) It helps to deploy applications on the cloud</li> </ol> | 1M |

|       |                                                                                                                                                                                                                                                                                     |    |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Q6.7  | The process by which the brain incrementally orders actions needed to complete a specific task is referred as _____<br>a) Planning problem<br>b) Partial order planning<br>c) Total order planning<br>d) Both Planning problem & Partial order planning                             | 1M |
| Q6.8  | Following is/are the components of the partial order planning.<br>a) Bindings<br>b) Goal<br>c) Causal Links<br>d) All of the mentioned                                                                                                                                              | 1M |
| Q 6.9 | Though local search algorithms are not systematic, key advantages would include _____<br>a) Less memory<br>b) More time<br>c) Finds a solution in large infinite space<br>d) Less memory & Finds a solution in large infinite space                                                 | 1M |
| Q6.10 | What are the main cons of hill-climbing search?<br>a) Terminates at local optimum & Does not find optimum solution<br>b) Terminates at global optimum & Does not find optimum solution<br>c) Does not find optimum solution & Fail to find a solution<br>d) Fail to find a solution | 1M |

Q. 6.1. Human brain uses: for reasoning: —

- Forward chaining
- backward chaining
- Resolution
- Predicate logic

1 m.



19/5/2022(E)

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**End Semester Examinations**  
**May-June 2022**

**Max. Marks: 100****Class: TY****Name of the Course: Digital Signal and Image Processing****Branch: Computer Engineering****Course Code: 2UCC601**

**Duration: 3 Hours**  
**Semester: VI**

**Instructions:**

- (1) **All Questions are Compulsory**
- (2) **Draw neat diagrams**
- (3) **Assume suitable data if necessary**

| Question No.  |                                                                                                                                                                                                                                                                                                                                                                      | Marks      |     |     |     |     |     |    |   |   |               |     |      |     |     |     |     |     |    |    |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|-----|-----|-----|-----|----|---|---|---------------|-----|------|-----|-----|-----|-----|-----|----|----|
| Q 1 (a)       | Classify the following DT systems on linearity, time invariance, causality<br>i. $y(n)=x^2(n)$<br>ii. $y(n)=e^{x(n)}$<br>iii. $y(n)=4x(n)+ x(n-3)$                                                                                                                                                                                                                   | 10         |     |     |     |     |     |    |   |   |               |     |      |     |     |     |     |     |    |    |
| Q 1 (b)       | i. What is Unitary transform matrix? Explain with example.<br>ii. Explain in short sampling and quantization method for digital image.                                                                                                                                                                                                                               | 10         |     |     |     |     |     |    |   |   |               |     |      |     |     |     |     |     |    |    |
| Q2 (a)        | Perform histogram equalization and draw new equalized histogram of the following image data<br><table border="1"><tr><td>Gray Level</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>No. of pixels</td><td>700</td><td>1000</td><td>950</td><td>600</td><td>300</td><td>250</td><td>105</td><td>95</td></tr></table> | Gray Level | 0   | 1   | 2   | 3   | 4   | 5  | 6 | 7 | No. of pixels | 700 | 1000 | 950 | 600 | 300 | 250 | 105 | 95 | 10 |
| Gray Level    | 0                                                                                                                                                                                                                                                                                                                                                                    | 1          | 2   | 3   | 4   | 5   | 6   | 7  |   |   |               |     |      |     |     |     |     |     |    |    |
| No. of pixels | 700                                                                                                                                                                                                                                                                                                                                                                  | 1000       | 950 | 600 | 300 | 250 | 105 | 95 |   |   |               |     |      |     |     |     |     |     |    |    |
| Q2 (b)        | Explain the use of high pass filter mask and high boost filter mask for spatial domain enhancement.                                                                                                                                                                                                                                                                  | 10         |     |     |     |     |     |    |   |   |               |     |      |     |     |     |     |     |    |    |
| Q3 (a)        | Compute DFT of the digital image given below using DIT-FFT algorithm (DIT- FFT signal flow graph/ Butterfly diagram)<br><table border="1"><tr><td>3</td><td>1</td><td>1</td><td>3</td></tr><tr><td>3</td><td>3</td><td>0</td><td>1</td></tr><tr><td>2</td><td>2</td><td>3</td><td>4</td></tr><tr><td>0</td><td>1</td><td>2</td><td>3</td></tr></table>               | 3          | 1   | 1   | 3   | 3   | 3   | 0  | 1 | 2 | 2             | 3   | 4    | 0   | 1   | 2   | 3   | 10  |    |    |
| 3             | 1                                                                                                                                                                                                                                                                                                                                                                    | 1          | 3   |     |     |     |     |    |   |   |               |     |      |     |     |     |     |     |    |    |
| 3             | 3                                                                                                                                                                                                                                                                                                                                                                    | 0          | 1   |     |     |     |     |    |   |   |               |     |      |     |     |     |     |     |    |    |
| 2             | 2                                                                                                                                                                                                                                                                                                                                                                    | 3          | 4   |     |     |     |     |    |   |   |               |     |      |     |     |     |     |     |    |    |
| 0             | 1                                                                                                                                                                                                                                                                                                                                                                    | 2          | 3   |     |     |     |     |    |   |   |               |     |      |     |     |     |     |     |    |    |

|           | <p align="center"><b>OR</b></p> <p>Design 4 x 4 Haar transform matrix and represent the row basis functions (basis images) graphically.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |             |    |     |    |     |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|----|-----|----|-----|----|-----|----|------|----|------|----|-----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| Q3 (b)    | Write 8x8 Hadamard transform matrix and its signal flow graph. Using the Butterfly diagram, compute Hadamard transform for<br>$x(n) = \{2, 3, 1, 3, 4, 3, 1, 1\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 10<br>10 |             |    |     |    |     |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| <b>OR</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |             |    |     |    |     |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| Q4 (a)    | Apply Region Splitting algorithm for the given image and the predicate is $[\max\{f(x, y)\} - \min\{f(x, y)\} \leq 4]$ , where $(x, y) \in R$ , where R is the given region. Show the quad Tree for the nodes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 10       |             |    |     |    |     |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|           | <table border="1"><tr><td>7</td><td>3</td><td>6</td><td>7</td><td>6</td><td>7</td><td>5</td><td>5</td></tr><tr><td>5</td><td>5</td><td>4</td><td>4</td><td>6</td><td>4</td><td>8</td><td>7</td></tr><tr><td>6</td><td>4</td><td>5</td><td>4</td><td>7</td><td>6</td><td>8</td><td>6</td></tr><tr><td>7</td><td>5</td><td>5</td><td>4</td><td>6</td><td>3</td><td>4</td><td>6</td></tr><tr><td>4</td><td>4</td><td>4</td><td>5</td><td>4</td><td>5</td><td>5</td><td>6</td></tr><tr><td>3</td><td>3</td><td>4</td><td>6</td><td>6</td><td>4</td><td>4</td><td>8</td></tr><tr><td>3</td><td>4</td><td>3</td><td>3</td><td>5</td><td>4</td><td>4</td><td>5</td></tr><tr><td>5</td><td>5</td><td>4</td><td>3</td><td>7</td><td>6</td><td>5</td><td>4</td></tr></table> | 7        | 3           | 6  | 7   | 6  | 7   | 5  | 5   | 5  | 5    | 4  | 4    | 6  | 4   | 8 | 7 | 6 | 4 | 5 | 4 | 7 | 6 | 8 | 6 | 7 | 5 | 5 | 4 | 6 | 3 | 4 | 6 | 4 | 4 | 4 | 5 | 4 | 5 | 5 | 6 | 3 | 3 | 4 | 6 | 6 | 4 | 4 | 8 | 3 | 4 | 3 | 3 | 5 | 4 | 4 | 5 | 5 | 5 | 4 | 3 | 7 | 6 | 5 | 4 |  |
| 7         | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 6        | 7           | 6  | 7   | 5  | 5   |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 5         | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4        | 4           | 6  | 4   | 8  | 7   |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 6         | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 5        | 4           | 7  | 6   | 8  | 6   |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 7         | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 5        | 4           | 6  | 3   | 4  | 6   |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 4         | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4        | 5           | 4  | 5   | 5  | 6   |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 3         | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4        | 6           | 6  | 4   | 4  | 8   |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 3         | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3        | 3           | 5  | 4   | 4  | 5   |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 5         | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4        | 3           | 7  | 6   | 5  | 4   |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|           | <p align="center"><b>OR</b></p> <p>What is image segmentation? Explain the following methods of image segmentation.</p> <ul style="list-style-type: none"><li>i. Region growing</li><li>ii. Split and Merge</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 10       |             |    |     |    |     |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| Q4 (b)    | Find variable length code using Huffman coding for the symbols and the data given below.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 10       |             |    |     |    |     |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|           | <table border="1"><thead><tr><th>Symbol</th><th>Probability</th></tr></thead><tbody><tr><td>a1</td><td>0.1</td></tr><tr><td>a2</td><td>0.2</td></tr><tr><td>a3</td><td>0.3</td></tr><tr><td>a4</td><td>0.14</td></tr><tr><td>a5</td><td>0.16</td></tr><tr><td>a6</td><td>0.1</td></tr></tbody></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Symbol   | Probability | a1 | 0.1 | a2 | 0.2 | a3 | 0.3 | a4 | 0.14 | a5 | 0.16 | a6 | 0.1 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| Symbol    | Probability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |             |    |     |    |     |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| a1        | 0.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |             |    |     |    |     |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| a2        | 0.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |             |    |     |    |     |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| a3        | 0.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |             |    |     |    |     |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| a4        | 0.14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |             |    |     |    |     |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| a5        | 0.16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |             |    |     |    |     |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| a6        | 0.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |             |    |     |    |     |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| Q5 (a)    | Write a detail note on (any one) <ul style="list-style-type: none"><li>i. Hough Transform</li><li>ii. Homomorphic filter</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 10       |             |    |     |    |     |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| Q5 (b)    | Explain in detail Hit or Miss Transform with an example                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 10       |             |    |     |    |     |    |     |    |      |    |      |    |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |

(10)

→ Q.3(b) Compute the discrete cosine transform (DCT) matrix for  $N=4$ .



26/05/2022(E)

**K. J. Somaiya College of Engineering, Mumbai-77**  
(Autonomous College Affiliated to University of Mumbai)

**End Semester Examinations**  
May-June 2022

**Max. Marks:** 100  
**Class:** TYBTech  
**Name of the Course:** Machine Learning  
**Branch:** Computer Engineering  
**Course Code:** 2UCE615

**Duration: 3 Hours**  
**Semester: VI**

**Instructions:**

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

| Question No. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks            |              |                  |              |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------|------------------|--------------|----------|------------|-----------|-----|------|------|-----|----|-------|-----|------|------|------|----|---|------|------|------|-----|-----|---|--------|--------|------|-----|-----|---|--------|-----|-----|-----|----|---|--------|-----|-----|------|-----|---|------|-----|-----|------|-----|---|-----|--------|------|-----|----|---|-----|-----|-----|-----|----|----|--------|--------|-----|-----|----|----|-----|--------|-----|------|-----|----|------|--------|------|------|-----|----|------|------|-----|-----|----|----|--------|--------|------|------|-----|-----|
| Q 1          | Solve any TWO (2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | (8)              |              |                  |              |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| (a)          | What are the steps in designing a Machine Learning Application?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (4)              |              |                  |              |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| (b)          | Define logit function. Explain the importance of logit function in logistic regression with appropriate example                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (4)              |              |                  |              |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| (c)          | Discuss the issues in Machine Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | (4)              |              |                  |              |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| Q2           | Solve any FOUR (4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | (32)             |              |                  |              |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| (a)          | <p>The sales of a company (in million dollars) for each year are shown in the table below.</p> <table><tr><td>X (year)</td><td>2005</td><td>2006</td><td>2007</td><td>2008</td><td>2009</td></tr><tr><td>Y (sales)</td><td>12</td><td>19</td><td>29</td><td>37</td><td>45</td></tr></table> <p>Find the least square regression line <math>y = ax + b</math>. Using that predict the sales for the year 2020 and 2021</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | X (year)         | 2005         | 2006             | 2007         | 2008     | 2009       | Y (sales) | 12  | 19   | 29   | 37  | 45 | (6+2) |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| X (year)     | 2005                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2006             | 2007         | 2008             | 2009         |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| Y (sales)    | 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 19               | 29           | 37               | 45           |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| (b)          | What is the goal of SVM? How is margin computed in SVM? What is the kernel trick in SVM?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | (2+3+3)          |              |                  |              |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| (c)          | <p>For the following data, construct a decision tree. Using entropy and information gain, determine which attribute is the root attribute.</p> <table><tr><th>Sr. No</th><th>Income</th><th>Defaulting Level</th><th>Credit Score</th><th>Location</th><th>Give Loan?</th></tr><tr><td>1</td><td>low</td><td>high</td><td>high</td><td>bad</td><td>no</td></tr><tr><td>2</td><td>low</td><td>high</td><td>high</td><td>good</td><td>no</td></tr><tr><td>3</td><td>high</td><td>high</td><td>high</td><td>bad</td><td>yes</td></tr><tr><td>4</td><td>medium</td><td>medium</td><td>high</td><td>bad</td><td>yes</td></tr><tr><td>5</td><td>medium</td><td>low</td><td>low</td><td>bad</td><td>no</td></tr><tr><td>6</td><td>medium</td><td>low</td><td>low</td><td>good</td><td>yes</td></tr><tr><td>7</td><td>high</td><td>low</td><td>low</td><td>good</td><td>Yes</td></tr><tr><td>8</td><td>low</td><td>medium</td><td>high</td><td>bad</td><td>No</td></tr><tr><td>9</td><td>low</td><td>low</td><td>low</td><td>bad</td><td>No</td></tr><tr><td>10</td><td>medium</td><td>medium</td><td>low</td><td>bad</td><td>No</td></tr><tr><td>11</td><td>low</td><td>medium</td><td>low</td><td>good</td><td>Yes</td></tr><tr><td>12</td><td>high</td><td>medium</td><td>high</td><td>good</td><td>yes</td></tr><tr><td>13</td><td>high</td><td>high</td><td>low</td><td>bad</td><td>no</td></tr><tr><td>14</td><td>medium</td><td>medium</td><td>high</td><td>good</td><td>yes</td></tr></table> | Sr. No           | Income       | Defaulting Level | Credit Score | Location | Give Loan? | 1         | low | high | high | bad | no | 2     | low | high | high | good | no | 3 | high | high | high | bad | yes | 4 | medium | medium | high | bad | yes | 5 | medium | low | low | bad | no | 6 | medium | low | low | good | yes | 7 | high | low | low | good | Yes | 8 | low | medium | high | bad | No | 9 | low | low | low | bad | No | 10 | medium | medium | low | bad | No | 11 | low | medium | low | good | Yes | 12 | high | medium | high | good | yes | 13 | high | high | low | bad | no | 14 | medium | medium | high | good | yes | (8) |
| Sr. No       | Income                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Defaulting Level | Credit Score | Location         | Give Loan?   |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| 1            | low                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | high             | high         | bad              | no           |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| 2            | low                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | high             | high         | good             | no           |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| 3            | high                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | high             | high         | bad              | yes          |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| 4            | medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | medium           | high         | bad              | yes          |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| 5            | medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | low              | low          | bad              | no           |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| 6            | medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | low              | low          | good             | yes          |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| 7            | high                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | low              | low          | good             | Yes          |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| 8            | low                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | medium           | high         | bad              | No           |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| 9            | low                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | low              | low          | bad              | No           |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| 10           | medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | medium           | low          | bad              | No           |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| 11           | low                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | medium           | low          | good             | Yes          |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| 12           | high                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | medium           | high         | good             | yes          |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| 13           | high                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | high             | low          | bad              | no           |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |
| 14           | medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | medium           | high         | good             | yes          |          |            |           |     |      |      |     |    |       |     |      |      |      |    |   |      |      |      |     |     |   |        |        |      |     |     |   |        |     |     |     |    |   |        |     |     |      |     |   |      |     |     |      |     |   |     |        |      |     |    |   |     |     |     |     |    |    |        |        |     |     |    |    |     |        |     |      |     |    |      |        |      |      |     |    |      |      |     |     |    |    |        |        |      |      |     |     |

|        |                                                                                                                                                                                                                                                                                                                             |       |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|        |                                                                                                                                                                                                                                                                                                                             |       |
| (d)    | Describe Hidden Markow Model with Example. List the issues of Hidden Markow Model.                                                                                                                                                                                                                                          | (2+6) |
| (e)    | Define Belief Network. Explain conditional Independence relation in Belief Network with an example.                                                                                                                                                                                                                         | (8)   |
| Q3     | Solve any TWO (2)                                                                                                                                                                                                                                                                                                           | (20)  |
| Q 3(a) | List down the steps of PCA algorithm.<br>Using PCA compute the transformed matrix of A where A is given by<br><div style="display: inline-block; border: 1px solid black; padding: 5px; margin: 10px;"> <math display="block">\begin{bmatrix} 4 &amp; 8 \\ 2 &amp; 6 \\ 0 &amp; 2 \\ -2 &amp; 1 \end{bmatrix}</math> </div> | (10)  |
| (b)    | What are the different processes under Feature Engineering? Explain each in brief.                                                                                                                                                                                                                                          | (10)  |
| (c)    | Why Singular Value Decomposition (SVD) is used? Explain SVD.                                                                                                                                                                                                                                                                | (10)  |
| Q 4    | Solve any TWO (2)                                                                                                                                                                                                                                                                                                           | (20)  |
| (a)    | For the given data [2, 7, 1, 9, 4], form 2 clusters using Expectation Maximization (EM) algorithm. Assume initial hypothesis $h = \langle \mu_1, \mu_2 \rangle = \langle 1, 2 \rangle$ , $\sigma = 1.0$ . Compute the new hypothesis. Perform 1st iteration only. Given,                                                    | (10)  |



|        | $E[Z_{ij}] = \frac{e^{(-1/(2\sigma^2)) \cdot (x_i - \mu_j)^2}}{\sum_{j=1}^n e^{(-1/(2\sigma^2)) \cdot (x_i - \mu_j)^2}}$ <p>Where,</p> <p><math>Z_{ij}</math> = expected value of hidden variable <math>h = \langle \mu_1, \mu_2 \rangle</math> hypothesis<br/> <math>i</math> = Data points<br/> <math>j</math> = clusters</p>                                                                                                                                                                                            |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|------|
| (b)    | <p>Use K-Means algorithm to create 2-clusters for given set of values:</p> <table border="1"> <tr> <th>Object</th> <th>A</th> <th>B</th> <th>C</th> <th>D</th> </tr> <tr> <td>X</td> <td>1</td> <td>2</td> <td>4</td> <td>5</td> </tr> <tr> <td>Y</td> <td>1</td> <td>1</td> <td>3</td> <td>4</td> </tr> </table>                                                                                                                                                                                                          | Object | A | B | C | D | X | 1 | 2 | 4 | 5 | Y | 1 | 1 | 3 | 4 | (10) |
| Object | A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | B      | C | D |   |   |   |   |   |   |   |   |   |   |   |   |      |
| X      | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2      | 4 | 5 |   |   |   |   |   |   |   |   |   |   |   |   |      |
| Y      | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1      | 3 | 4 |   |   |   |   |   |   |   |   |   |   |   |   |      |
| (c)    | Can a nonlinearly separable pattern become separable using RBF? How is it made separable? Explain with suitable example                                                                                                                                                                                                                                                                                                                                                                                                    | (10)   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |
| Q5     | Solve any TWO (2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | (20)   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |
| (a)    | How reinforcement learning is different from Supervised Learning? Take a suitable example to explain the working of Reinforcement Learning Model.                                                                                                                                                                                                                                                                                                                                                                          | (10)   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |
| (b)    | With suitable equations and graphs, explain the types of activation functions used in neural networks.                                                                                                                                                                                                                                                                                                                                                                                                                     | (10)   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |
| (c)    | <p>Determine the weights after four steps of training for perceptron learning rule of a single layer neural network starting with initial weights-<br/> <math>W = [0 \ 0]^T</math>, inputs as <math>X_1 = [2 \ 2]^T</math>, <math>X_2 = [1 \ -2]^T</math>, <math>X_3 = [-2 \ 2]^T</math>, <math>X_4 = [-1 \ 1]^T</math>,<br/> <math>d_1 = 0</math>, <math>d_2 = 1</math>, <math>d_3 = 0</math>, <math>d_4 = 1</math> and <math>c = 1</math> Where <math>c</math> is learning rate and <math>d</math> is desired output</p> | (10)   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |

Pg 3 of 3

26/05/2022 (E)

K. J. Somaiya College of Engineering, Mumbai-77  
(Autonomous College Affiliated to University of Mumbai)

End Semester Examinations  
May-June 2022

Max. Marks: 100

Class: **TY B Tech**

Name of the Course: Software Testing and Quality Assurance

Course Code: 2UCE613

Duration: 3 hrs.

Semester: VI

Branch: Comp

**Instructions:**

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

| Question No. |                                                                                                                                                                                                                                                                                                                                       | Marks            |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Q 1 (a)      | Define the terms<br>i. Failure<br>ii. Error<br>iii. Fault<br>iv. Verification<br>v. Validation                                                                                                                                                                                                                                        | 05               |
| Q 1 (b)      | Explain different views of quality. Support your answer with example<br><br><b>OR</b><br><br>Explain different debugging techniques and debugging heuristic.                                                                                                                                                                          | 10<br><br><br>10 |
| Q 1 (c)      | Explain different test levels and classify them as verification or validation                                                                                                                                                                                                                                                         | 05               |
| Q 2 (a)      | Explain different levels of testing and classify each as verification or validation.                                                                                                                                                                                                                                                  | 10               |
| Q 2 (b)      | Write an algorithm to find maximum of three numbers, convert the algorithm into control flow graph and design test cases for the same<br><br><b>OR</b><br><b>Process Design Language</b><br>Convert the PDL into CFG and find cyclomatic complexity<br>If (c1 or c2) s1;<br>Else if (c3) s2;<br>Else s3;<br><b>Control Flow Graph</b> | 10<br><br><br>10 |
| Q 3 (a)      | Explain Alpha, Beta and UAT ( <b>Users Acceptance Test</b> )<br><br><b>OR</b><br><br>Explain different types of integration testing and the need of Stub and Driver modules                                                                                                                                                           | 10<br><br><br>10 |
| Q 3 (b)      | What is black box testing? Explain Boundary value and Equivalence partitioning with example.                                                                                                                                                                                                                                          | 10               |



|        |                                                                       |    |
|--------|-----------------------------------------------------------------------|----|
| Q4 (a) | Explain different types of system tests                               | 10 |
|        | OR                                                                    |    |
|        | Write short note on<br>i. Regression tests<br>ii. Documentation tests | 10 |
| Q4 (b) | Explain McCall's quality factors                                      | 10 |
| Q5 (a) | Write note on ISO 9126 quality characteristics                        | 10 |
| Q5 (b) | On what principles ISO 9000:2000 is based                             | 10 |

26/05/2022(E)

**K. J. Somaiya College of Engineering, Mumbai-77**  
(Autonomous College Affiliated to University of Mumbai)

**End Semester Examinations**  
May-June 2022

**Max. Marks: 100**

Class: TY B Tech

Name of the Course: Micro Services and DevOps

Course Code: 2UCE617

**Duration: 3 Hour**

Semester: VI

Branch: COMP

**Instructions:**

- (1) **All Questions are Compulsory**
- (2) **Draw neat diagrams**
- (3) **Assume suitable data if necessary**

| Question No. |                                                                                                                                                                                                                                                                                                | Marks |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Q 1 (a)      | With respect to DevOps what is people, processes and tools? Explain each one with an example.                                                                                                                                                                                                  | 10    |
| Q 1 (b)      | What are DevOps focus areas? Explain any two in detail.<br><br><b>OR</b><br><br>Explain the working of DevOps                                                                                                                                                                                  | 10    |
| Q2 (a)       | Explain Configuration Management in DevOps.                                                                                                                                                                                                                                                    | 10    |
| Q2 (b)       | Explain Continuous Integration and Continuous Delivery in DevOps.<br><br><b>OR</b><br><br>What are DevOps Principles and Practices? What is the role of DevOps Engineer?                                                                                                                       | 10    |
| Q3 (a)       | Describe the steps to develop a pipeline in Jenkins to automatically deploy web application on web server.                                                                                                                                                                                     | 10    |
| Q3 (b)       | What are different tools to build pipeline? How to start building pipeline? Explain steps involved in DevOps Pipeline Implementation.<br><br><b>OR</b><br><br>What is a Deployment Pipeline? What are the main stages of a Deployment Pipeline? What are the benefits of a Deployment Pipeline | 10    |



|        |                                                                                                                                                                                                                                                                                                                                 |    |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Q4 (a) | <p>What are the pros and cons of MicroService Architecture? Explain</p> <p style="text-align: center;"><b>OR</b></p> <p>What are the design patterns of MicroServices? Explain any two in detail.</p>                                                                                                                           | 10 |
| Q4 (b) | <p>What is Cohesion and coupling in Microservices? Explain MicroServices implementation steps <i>with suitable examples.</i></p>                                                                                                                                                                                                | 10 |
| Q5 (a) | <p>What are Spring Cloud Netflix OSS Tools? Explain the use of any two tools with the help of an example.</p>                                                                                                                                                                                                                   | 10 |
| Q5 (b) | <p>How to examine the following</p> <ol style="list-style-type: none"> <li>1. POM File</li> <li>2. Application.properties</li> <li>3. ApplicationTests.java</li> </ol> <p style="text-align: center;"><b>OR</b></p> <p>What is Spring Boot and Spring Cloud? Why is Spring Cloud used? Discuss the features of Spring Cloud</p> | 10 |

**K. J. Somaiya College of Engineering, Mumbai-77**  
(Autonomous College Affiliated to University of Mumbai)

**End Semester Examinations**

May-June 2022

**Max. Marks: 100**

**Class: TY**

**Name of the Course: Artificial Intelligence**

**Course Code: 2UCC603**

**Duration: 3 hours**

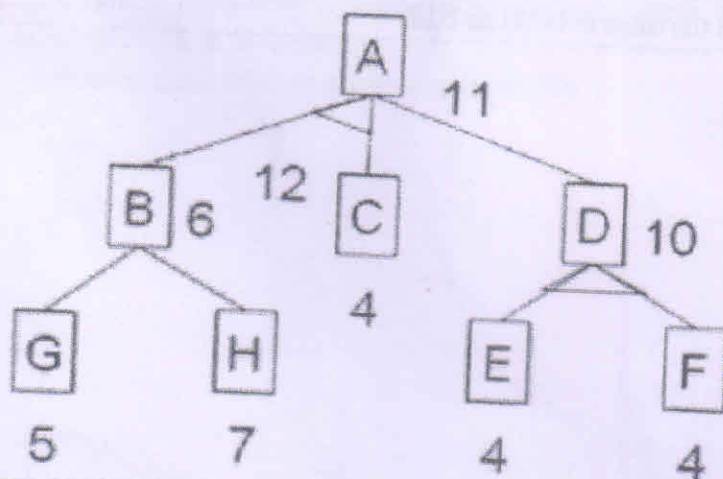
**Semester: VI**

**Branch: Computer**

**Instructions:**

- (1) **All Questions are Compulsory**
- (2) **Draw neat diagrams**
- (3) **Assume suitable data if necessary**

| Question No. |                                                                                                                                                                                                                                                         | Marks               |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Q 1 (a)      | State and explain the "AI Problems". Brief on Mundane, Formal and Expert tasks.<br><br>OR<br>(i) List 5 things AI cannot do yet.<br>(ii) List 5 applications of AI in daily life.                                                                       | 10M<br><br>5M<br>5M |
| Q 1 (b)      | Describe each of the components of the Utility based agent. Support with a neat diagram.<br>Comment on the example of a Mars lander on the surface of mars with obstacles in its way and how would it be different in a Goal based agent with learning. | 3M<br>3M<br>4M      |
| Q2 (a)       | What are Production rules?<br>Enumerate the steps to problem formulation in Block's world problem.                                                                                                                                                      | 02M<br>08M          |
| Q2 (b)       | Define straight line distance as a heuristic.<br>Distinguish between AO* Search Algorithm and A* search algorithm.<br>Solve using AO* search algorithm                                                                                                  | 02M<br>03M<br>05M   |





|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Q3 (a) | Solve using CSP and show all steps clearly.<br>TWO + TWO = FOUR <b>OR 3A in Addendum</b>                                                                                                                                                                                                                                                                                                                                                                                                  | 10M |
| Q3 (b) | Explain the steps of knowledge engineering process for a detective agent.<br>Give sample output for every step.                                                                                                                                                                                                                                                                                                                                                                           | 10M |
| Q4 (a) | Brief on the following<br>(i) Universal Generalization<br>(ii) Universal Instantiation.<br>(iii) Existential Instantiation<br>(iv) Existential Introduction. <b>OR 4A in Addendum</b>                                                                                                                                                                                                                                                                                                     | 10M |
| Q4 (b) | Translate the following set of sentences into FOL. Use backward chaining on the goal- <u>Joel is at museum</u> .<br><br>a. Joel is a bulldog.<br>b. John is Joel's master.<br>c. The day is Saturday<br>d. It is cold on Saturday.<br>e. Joel is trained.<br>f. Trained bulldogs are good dogs.<br>g. If a dog is a good dog and has a master and at some place, then he will be with his master.<br>h. If the day is Saturday and the day is cold, John is at the museum.<br><b>Give</b> | 10M |
| Q5 (a) | STRIPS solution for the problem of changing flat tire with a spare one in car trunk.<br>Clearly specify initial state, goal(s), actions and their pre-post conditions                                                                                                                                                                                                                                                                                                                     | 10M |
| Q5 (b) | Differentiate between Supervised learning, Unsupervised learning, Semi supervised learning and reinforcement learning.<br><b>OR</b><br>List and explain the steps to build an NLP                                                                                                                                                                                                                                                                                                         | 10M |

### Addendum to the question paper.

**K. J. Somaiya College of Engineering, Mumbai-77**  
(Autonomous College Affiliated to University of Mumbai)

**End Semester Exam**  
**MAY-JUNE 2022**

Class: TY B.Tech  
Name of the Course: Artificial Intelligence  
Course Code: 2UCC603

Semester: VI  
Branch: Computer Engg.

Q. 3 A. Consider an instance of 0/1 knapsack as:

$N=6$ , Capacity = 20, weight =  $\{2,4,5,3,4,2\}$  and profit =  $\{10,15,15,20,200,50\}$

Solve the problem using genetic algorithms.

- Clearly state thresholds or any other assumptions
- Show step by step output after every step.

**Compute solution upto only ONE generation.**

Q. 4 A Consider the Lung Cancer Problem- Pollution and smoking are the factors those affect a patient's chance of having cancer. Similarly, having cancer affects the patient's breathing and the chances of having a positive X-ray result.

- Draw Bayesian network for the above stated problem 2M
- Assign probabilities to every node in the network 2M
- Compute probability of a smoker having lung cancer, has a positive X-ray result but faces no problem in breathing. 3M
- State reasons of uncertainty in general. 3M



26/05/2022(E)

**K. J. Somaiya College of Engineering, Mumbai-77**  
(Autonomous College Affiliated to University of Mumbai)

**End Semester Examinations**

May-June 2022

**Max. Marks: 100**

Class: T Y B.Tech

Name of the Course: **Data Mining and Business Intelligence**

Branch: Computer

Course Code: 2UCE602

**Duration: 3 hrs.**

Semester: VI

**Instructions:**

- (1) **All Questions are Compulsory**
- (2) **Draw neat diagrams**
- (3) **Assume suitable data if necessary**

| Question No. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Q 1 (a)      | <p>i) Suppose that you are employed as a data mining consultant for an Somaiya Vidhyavihar University. Describe how data mining can help the university by giving specific examples of any 2 data mining techniques that can be applied.</p> <p>ii) With diagram, explain the architecture of typical data mining system you would choose for the given above case study? What is purpose of each component?</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 10    |
| Q 1 (b)      | <p>i) Suppose the dataset for analysis includes the attributes - frequency of stop words in documents. The values given are: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.<br/>Use the smoothing by bin means with a depth of 3.<br/>Give the five-box summary of the data.</p> <p>ii) Consider the sample dataset of loan dispersed by Peer to peer lending firm. Discuss any 2 better strategies to handle missing values. The description of columns in the dataset is as follows.<br/>Total Loan Repayments- Total number of loans previously paid by the borrower on the platform,<br/>Credit Score- Credit score of the borrower of the loan,<br/>State- State in which the loan was issued,<br/>Monthly EMI- Monthly EMI paid by the borrower,<br/>Monthly Income- Monthly income of the borrower,<br/>Liquidity ratio- Monthly EMI/ Monthly Income,<br/>Defaulted- If the borrower defaulted on the loan (dependent variable)</p> | 10    |

|                       | <table><tr><th>Total loan repayments</th><th>Credit Score</th><th>State</th><th>Monthly EMI</th><th>Monthly Income</th><th>Liquidity ratio</th><th>Defaulted</th></tr><tr><td>0</td><td>680</td><td>AP</td><td>15000</td><td>32000</td><td>High</td><td>Y</td></tr><tr><td>1</td><td>600</td><td>GUJ</td><td>16000</td><td>20000</td><td>Very High</td><td>N</td></tr><tr><td>0</td><td>700</td><td>GUJ</td><td>9000</td><td>45000</td><td>High</td><td>N</td></tr><tr><td>2</td><td>720</td><td>MAH</td><td>4000</td><td>34000</td><td>Low</td><td>N</td></tr><tr><td>0</td><td>650</td><td></td><td>23000</td><td>33000</td><td>High</td><td>Y</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Total loan repayments | Credit Score           | State          | Monthly EMI     | Monthly Income | Liquidity ratio | Defaulted | 0    | 680 | AP | 15000 | 32000 | High | Y   | 1  | 600 | GUJ | 16000 | 20000 | Very High | N  | 0  | 700 | GUJ                  | 9000 | 45000 | High | N     | 2   | 720   | MAH | 4000  | 34000 | Low  | N   | 0     | 650 |  | 23000 | 33000 | High | Y |  |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------------|----------------|-----------------|----------------|-----------------|-----------|------|-----|----|-------|-------|------|-----|----|-----|-----|-------|-------|-----------|----|----|-----|----------------------|------|-------|------|-------|-----|-------|-----|-------|-------|------|-----|-------|-----|--|-------|-------|------|---|--|
| Total loan repayments | Credit Score                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | State                 | Monthly EMI            | Monthly Income | Liquidity ratio | Defaulted      |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 0                     | 680                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | AP                    | 15000                  | 32000          | High            | Y              |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 1                     | 600                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | GUJ                   | 16000                  | 20000          | Very High       | N              |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 0                     | 700                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | GUJ                   | 9000                   | 45000          | High            | N              |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 2                     | 720                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | MAH                   | 4000                   | 34000          | Low             | N              |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 0                     | 650                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       | 23000                  | 33000          | High            | Y              |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| Q2 (a)                | <p>The following table shows the data collected by a state highway patrol safety division on stopping distances.</p> <table><tr><th>Speed(mph)</th><th>Stopping Distance (ft)</th></tr><tr><td>12</td><td>17.5</td></tr><tr><td>15</td><td>28</td></tr><tr><td>20</td><td>41.5</td></tr><tr><td>25</td><td>56</td></tr><tr><td>32</td><td>77.5</td></tr><tr><td>37</td><td>104</td></tr><tr><td>42</td><td>122</td></tr><tr><td>47</td><td>158</td></tr><tr><td>52</td><td>177.5</td></tr><tr><td>32</td><td>80</td></tr></table> <p>Use the method of least squares to formulate an equation for the prediction of stopping distance of a vehicle according the speed traveled.<br/>Predict the stopping distance for a vehicle travelling at 48 mph.</p> <p style="text-align: center;"><b>OR</b></p> <p>i. Suppose a bank would like to develop a classifier that guards against fraudulent credit card transactions. Discuss different strategies as to how to induce a quality classifier based on a large set of non-fraudulent examples and a very small set of fraudulent cases.</p> <p>ii. You are given 10 training samples. They are divided into 4 classes: A,B,C,D. One sample belongs to A, two to B, three belong to C, and four belong to D. Use the following table to answer the following questions:</p> <table><tr><th>p</th><th>Log<sub>2</sub>(p)</th></tr><tr><td>0.1</td><td>-3.32</td></tr><tr><td>0.2</td><td>-2.32</td></tr><tr><td>0.3</td><td>-1.74</td></tr><tr><td>0.4</td><td>-1.32</td></tr><tr><td>0.5</td><td>-1.0</td></tr><tr><td>0.6</td><td>-0.74</td></tr></table> <p>What is the total information contained in the samples?<br/>What is the total Gini index?</p> | Speed(mph)            | Stopping Distance (ft) | 12             | 17.5            | 15             | 28              | 20        | 41.5 | 25  | 56 | 32    | 77.5  | 37   | 104 | 42 | 122 | 47  | 158   | 52    | 177.5     | 32 | 80 | p   | Log <sub>2</sub> (p) | 0.1  | -3.32 | 0.2  | -2.32 | 0.3 | -1.74 | 0.4 | -1.32 | 0.5   | -1.0 | 0.6 | -0.74 | 10  |  |       |       |      |   |  |
| Speed(mph)            | Stopping Distance (ft)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                       |                        |                |                 |                |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 12                    | 17.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                       |                        |                |                 |                |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 15                    | 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                       |                        |                |                 |                |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 20                    | 41.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                       |                        |                |                 |                |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 25                    | 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                       |                        |                |                 |                |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 32                    | 77.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                       |                        |                |                 |                |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 37                    | 104                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       |                        |                |                 |                |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 42                    | 122                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       |                        |                |                 |                |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 47                    | 158                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       |                        |                |                 |                |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 52                    | 177.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |                        |                |                 |                |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 32                    | 80                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                       |                        |                |                 |                |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| p                     | Log <sub>2</sub> (p)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                       |                        |                |                 |                |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 0.1                   | -3.32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |                        |                |                 |                |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 0.2                   | -2.32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |                        |                |                 |                |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 0.3                   | -1.74                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |                        |                |                 |                |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 0.4                   | -1.32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |                        |                |                 |                |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 0.5                   | -1.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                       |                        |                |                 |                |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |
| 0.6                   | -0.74                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |                        |                |                 |                |                 |           |      |     |    |       |       |      |     |    |     |     |       |       |           |    |    |     |                      |      |       |      |       |     |       |     |       |       |      |     |       |     |  |       |       |      |   |  |



| Q2 (b) | <p>Explain the steps of the ID3 algorithm for generating Decision trees.</p> <p style="text-align: center;"><b>OR</b></p> <p>Imagine that you created a large and complex decision tree from a training dataset that contain many attributes. Your tree has a high accuracy on the training set, but the accuracy on the test set is very low. Explain why the accuracy on the test data could be so much lower than the training data. Discuss briefly methods to improve your decision tree accuracy on test data set</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 10          |             |             |             |             |             |             |    |    |    |   |             |             |             |             |             |             |            |    |  |   |             |             |             |             |             |             |    |  |  |   |             |            |            |             |             |    |  |  |  |   |             |             |             |            |    |  |  |  |  |   |            |             |             |    |  |  |  |  |  |   |             |             |    |  |  |  |  |  |  |   |             |    |  |  |  |  |  |  |  |   |    |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|----|----|----|---|-------------|-------------|-------------|-------------|-------------|-------------|------------|----|--|---|-------------|-------------|-------------|-------------|-------------|-------------|----|--|--|---|-------------|------------|------------|-------------|-------------|----|--|--|--|---|-------------|-------------|-------------|------------|----|--|--|--|--|---|------------|-------------|-------------|----|--|--|--|--|--|---|-------------|-------------|----|--|--|--|--|--|--|---|-------------|----|--|--|--|--|--|--|--|---|----|
| Q3 (a) | <p>Consider the 2-D points and Euclidean distance matrix given below.<br/>A1=(2,10), A2=(2,5), A3=(8,4), A4=(5,8), A5=(7,5), A6=(6,4), A7=(1,2) ,<br/>A8=(4,9)</p> <table border="1"><thead><tr><th></th><th>A1</th><th>A2</th><th>A3</th><th>A4</th><th>A5</th><th>A6</th><th>A7</th><th>A8</th></tr></thead><tbody><tr><th>A1</th><td>0</td><td><math>\sqrt{25}</math></td><td><math>\sqrt{36}</math></td><td><math>\sqrt{13}</math></td><td><math>\sqrt{50}</math></td><td><math>\sqrt{52}</math></td><td><math>\sqrt{65}</math></td><td><math>\sqrt{5}</math></td></tr><tr><th>A2</th><td></td><td>0</td><td><math>\sqrt{37}</math></td><td><math>\sqrt{18}</math></td><td><math>\sqrt{25}</math></td><td><math>\sqrt{17}</math></td><td><math>\sqrt{10}</math></td><td><math>\sqrt{20}</math></td></tr><tr><th>A3</th><td></td><td></td><td>0</td><td><math>\sqrt{25}</math></td><td><math>\sqrt{2}</math></td><td><math>\sqrt{2}</math></td><td><math>\sqrt{53}</math></td><td><math>\sqrt{41}</math></td></tr><tr><th>A4</th><td></td><td></td><td></td><td>0</td><td><math>\sqrt{13}</math></td><td><math>\sqrt{17}</math></td><td><math>\sqrt{52}</math></td><td><math>\sqrt{2}</math></td></tr><tr><th>A5</th><td></td><td></td><td></td><td></td><td>0</td><td><math>\sqrt{2}</math></td><td><math>\sqrt{45}</math></td><td><math>\sqrt{25}</math></td></tr><tr><th>A6</th><td></td><td></td><td></td><td></td><td></td><td>0</td><td><math>\sqrt{29}</math></td><td><math>\sqrt{29}</math></td></tr><tr><th>A7</th><td></td><td></td><td></td><td></td><td></td><td></td><td>0</td><td><math>\sqrt{58}</math></td></tr><tr><th>A8</th><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>0</td></tr></tbody></table> <p>Use the hierarchical agglomerative clustering on the above 8 points and draw dendrograms, clearly showing the relevant distance information using Single link, Complete link.</p> |             | A1          | A2          | A3          | A4          | A5          | A6          | A7 | A8 | A1 | 0 | $\sqrt{25}$ | $\sqrt{36}$ | $\sqrt{13}$ | $\sqrt{50}$ | $\sqrt{52}$ | $\sqrt{65}$ | $\sqrt{5}$ | A2 |  | 0 | $\sqrt{37}$ | $\sqrt{18}$ | $\sqrt{25}$ | $\sqrt{17}$ | $\sqrt{10}$ | $\sqrt{20}$ | A3 |  |  | 0 | $\sqrt{25}$ | $\sqrt{2}$ | $\sqrt{2}$ | $\sqrt{53}$ | $\sqrt{41}$ | A4 |  |  |  | 0 | $\sqrt{13}$ | $\sqrt{17}$ | $\sqrt{52}$ | $\sqrt{2}$ | A5 |  |  |  |  | 0 | $\sqrt{2}$ | $\sqrt{45}$ | $\sqrt{25}$ | A6 |  |  |  |  |  | 0 | $\sqrt{29}$ | $\sqrt{29}$ | A7 |  |  |  |  |  |  | 0 | $\sqrt{58}$ | A8 |  |  |  |  |  |  |  | 0 | 10 |
|        | A1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | A2          | A3          | A4          | A5          | A6          | A7          | A8          |    |    |    |   |             |             |             |             |             |             |            |    |  |   |             |             |             |             |             |             |    |  |  |   |             |            |            |             |             |    |  |  |  |   |             |             |             |            |    |  |  |  |  |   |            |             |             |    |  |  |  |  |  |   |             |             |    |  |  |  |  |  |  |   |             |    |  |  |  |  |  |  |  |   |    |
| A1     | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\sqrt{25}$ | $\sqrt{36}$ | $\sqrt{13}$ | $\sqrt{50}$ | $\sqrt{52}$ | $\sqrt{65}$ | $\sqrt{5}$  |    |    |    |   |             |             |             |             |             |             |            |    |  |   |             |             |             |             |             |             |    |  |  |   |             |            |            |             |             |    |  |  |  |   |             |             |             |            |    |  |  |  |  |   |            |             |             |    |  |  |  |  |  |   |             |             |    |  |  |  |  |  |  |   |             |    |  |  |  |  |  |  |  |   |    |
| A2     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0           | $\sqrt{37}$ | $\sqrt{18}$ | $\sqrt{25}$ | $\sqrt{17}$ | $\sqrt{10}$ | $\sqrt{20}$ |    |    |    |   |             |             |             |             |             |             |            |    |  |   |             |             |             |             |             |             |    |  |  |   |             |            |            |             |             |    |  |  |  |   |             |             |             |            |    |  |  |  |  |   |            |             |             |    |  |  |  |  |  |   |             |             |    |  |  |  |  |  |  |   |             |    |  |  |  |  |  |  |  |   |    |
| A3     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             | 0           | $\sqrt{25}$ | $\sqrt{2}$  | $\sqrt{2}$  | $\sqrt{53}$ | $\sqrt{41}$ |    |    |    |   |             |             |             |             |             |             |            |    |  |   |             |             |             |             |             |             |    |  |  |   |             |            |            |             |             |    |  |  |  |   |             |             |             |            |    |  |  |  |  |   |            |             |             |    |  |  |  |  |  |   |             |             |    |  |  |  |  |  |  |   |             |    |  |  |  |  |  |  |  |   |    |
| A4     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |             | 0           | $\sqrt{13}$ | $\sqrt{17}$ | $\sqrt{52}$ | $\sqrt{2}$  |    |    |    |   |             |             |             |             |             |             |            |    |  |   |             |             |             |             |             |             |    |  |  |   |             |            |            |             |             |    |  |  |  |   |             |             |             |            |    |  |  |  |  |   |            |             |             |    |  |  |  |  |  |   |             |             |    |  |  |  |  |  |  |   |             |    |  |  |  |  |  |  |  |   |    |
| A5     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |             |             | 0           | $\sqrt{2}$  | $\sqrt{45}$ | $\sqrt{25}$ |    |    |    |   |             |             |             |             |             |             |            |    |  |   |             |             |             |             |             |             |    |  |  |   |             |            |            |             |             |    |  |  |  |   |             |             |             |            |    |  |  |  |  |   |            |             |             |    |  |  |  |  |  |   |             |             |    |  |  |  |  |  |  |   |             |    |  |  |  |  |  |  |  |   |    |
| A6     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |             |             |             | 0           | $\sqrt{29}$ | $\sqrt{29}$ |    |    |    |   |             |             |             |             |             |             |            |    |  |   |             |             |             |             |             |             |    |  |  |   |             |            |            |             |             |    |  |  |  |   |             |             |             |            |    |  |  |  |  |   |            |             |             |    |  |  |  |  |  |   |             |             |    |  |  |  |  |  |  |   |             |    |  |  |  |  |  |  |  |   |    |
| A7     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |             |             |             |             | 0           | $\sqrt{58}$ |    |    |    |   |             |             |             |             |             |             |            |    |  |   |             |             |             |             |             |             |    |  |  |   |             |            |            |             |             |    |  |  |  |   |             |             |             |            |    |  |  |  |  |   |            |             |             |    |  |  |  |  |  |   |             |             |    |  |  |  |  |  |  |   |             |    |  |  |  |  |  |  |  |   |    |
| A8     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |             |             |             |             |             | 0           |    |    |    |   |             |             |             |             |             |             |            |    |  |   |             |             |             |             |             |             |    |  |  |   |             |            |            |             |             |    |  |  |  |   |             |             |             |            |    |  |  |  |  |   |            |             |             |    |  |  |  |  |  |   |             |             |    |  |  |  |  |  |  |   |             |    |  |  |  |  |  |  |  |   |    |
| Q3 (b) | <p>Suppose that we have the following data:<br/>A(2, 0), B(1,2), C(2,2), D(3,2), E(2,3), F(3,3), G(2,4), H(3,4), I(4,4), J(3,5)<br/>Identify the cluster by applying the k-means algorithm, with k=2. Try using initial cluster centers as far apart as possible. Calculate the quality of the clusters</p> <p style="text-align: center;"><b>OR</b></p> <p>i) K-means and K-mediods are similar techniques. In what way using mediod instead of mean change clustering. Which of these methods are preferred?</p> <p>ii) Discuss the strength and weakness of BIRCH clustering algorithm.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 10          |             |             |             |             |             |             |    |    |    |   |             |             |             |             |             |             |            |    |  |   |             |             |             |             |             |             |    |  |  |   |             |            |            |             |             |    |  |  |  |   |             |             |             |            |    |  |  |  |  |   |            |             |             |    |  |  |  |  |  |   |             |             |    |  |  |  |  |  |  |   |             |    |  |  |  |  |  |  |  |   |    |
| Q4 (a) | <p>How apriori property is used in apriori algorithm ?What are limitation of apriori algorithm. Discuss any 2 approaches to improve the efficiency of apriori algorithm</p> <p style="text-align: center;"><b>OR</b></p> <p>How do we use different variations of apriori algorithm for mining multilevel association rules.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 10          |             |             |             |             |             |             |    |    |    |   |             |             |             |             |             |             |            |    |  |   |             |             |             |             |             |             |    |  |  |   |             |            |            |             |             |    |  |  |  |   |             |             |             |            |    |  |  |  |  |   |            |             |             |    |  |  |  |  |  |   |             |             |    |  |  |  |  |  |  |   |             |    |  |  |  |  |  |  |  |   |    |

| Q4 (b) | <p>Given the minimum support is 2 and minimum confidence is 70%. Find frequent item set and strong association rules using mining with vertical data format algorithm.</p> <table><tr><th>TID</th><th>Items in transaction</th></tr><tr><td>1</td><td>A,B,C</td></tr><tr><td>2</td><td>B,C,D,E</td></tr><tr><td>3</td><td>C,D</td></tr><tr><td>4</td><td>A,B,D</td></tr><tr><td>5</td><td>A,B,C</td></tr></table> | TID | Items in transaction | 1 | A,B,C | 2 | B,C,D,E | 3 | C,D | 4 | A,B,D | 5 | A,B,C | 10 |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------------------|---|-------|---|---------|---|-----|---|-------|---|-------|----|
| TID    | Items in transaction                                                                                                                                                                                                                                                                                                                                                                                              |     |                      |   |       |   |         |   |     |   |       |   |       |    |
| 1      | A,B,C                                                                                                                                                                                                                                                                                                                                                                                                             |     |                      |   |       |   |         |   |     |   |       |   |       |    |
| 2      | B,C,D,E                                                                                                                                                                                                                                                                                                                                                                                                           |     |                      |   |       |   |         |   |     |   |       |   |       |    |
| 3      | C,D                                                                                                                                                                                                                                                                                                                                                                                                               |     |                      |   |       |   |         |   |     |   |       |   |       |    |
| 4      | A,B,D                                                                                                                                                                                                                                                                                                                                                                                                             |     |                      |   |       |   |         |   |     |   |       |   |       |    |
| 5      | A,B,C                                                                                                                                                                                                                                                                                                                                                                                                             |     |                      |   |       |   |         |   |     |   |       |   |       |    |
| Q5 (a) | <p>What is need of Business Intelligence? With suitable diagram, discuss the three main components of Business Intelligence architecture</p>                                                                                                                                                                                                                                                                      | 10  |                      |   |       |   |         |   |     |   |       |   |       |    |
| Q5 (b) | <p>State the areas where Business intelligence and data mining can be used for Banking and Finance.</p> <p style="text-align: center;"><b>OR</b></p> <p>State the areas where Business intelligence and data mining can be used for Click stream mining, Telecommunication industry</p>                                                                                                                           | 10  |                      |   |       |   |         |   |     |   |       |   |       |    |